

Master Compilation Report: Cerebellar Reservoir Optimization & Empirical Validation

End-to-End Synthesis of Every Pipeline Stage, Mathematical Change, and Empirical
Benchmark

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Abstract

This master compilation report provides a complete, rigorous synthesis of every development stage, mathematical re-formulation, biophysical modification, and empirical benchmark conducted across the **ExactRModel** Cerebellar Reservoir architecture. Covering pre-training optimization, unbounded logit manifold calibration, adversarial audit of the ontogenetic prior hypothesis, Tsodyks-Markram biophysics, generator benchmarks, discrete-to-continuous 2AFC reversal task ingestion (16,029 human trials across 128 participants), discrete cognitive baselines (EV-RW vs. WSLs), Stage 6 LOOCV predictive cross-validation, and Stage 7 multiplexed kinematics with the Drift Diffusion Model (DDM) bridge, this document details the exact quantitative effect of every architectural change.

1 Master Progression of Pipeline Stages & Quantitative Effects

Table 1 provides an executive summary of every project stage, the exact intervention applied, and the quantitative effect on network performance.

Table 1: Master Progression Across All Pipeline Stages & Effects of Interventions.

Pipeline Stage	Core Objective	Architectural Intervention	Quantitative Effect / Result
1. GP Surrogate Calibration	Map 10D hyperparameter space ($N = 100$ Maximin LHS)	Fitted Matérn 5/2 Gaussian Process Surrogate Model	Replaced expensive network simulations with fast differentiable $U_{\text{marginal}}(\Theta)$ ($R^2 > 0.95$).
2. Unbounded Logit Manifold	Eliminate parameter boundary collapse (BCR = 60%)	Unbounded Latent Space $\tilde{\Theta} \in \mathbb{R}^5$ via Inverse Sigmoidal Projection	Analytical BCR = 0.0000 (0% Clamping). $R^2(D_{fb}) = 0.935$, $R^2(\rho) = 0.960$. $\lambda_{fb}(0.20) = 0.9783 \leq 0.99$.
3. Adversarial Prior Audit	Audit Kuramoto pre-training against empirical human shocks	Formalized 3 Failure Modes (Driven Chaos, Rank Collapse, Stiffening)	Calculated Empirical Safety Metric $\mathcal{R} = 58.4/100$ (High Vulnerability Risk).
4. Stage 1: C++ Bio-physics	Test Tsodyks-Markram STD D_t under DC shock (+10.0)	Integrated dynamic depression differential equation $\dot{D}_t$	Reduced max $ h_{\text{pre}} $ from 30.81 to 10.93; Saturation dropped from 70.13% to 0.00% .
5. Stage 2: Generator Deep-Dive	Compare Pure Sines (Model A) vs Broadband (Model B)	Evaluated held-out stationary empirical non-noise stream	Model A degraded by -550.83% ($\lambda_{\text{driven}} = 0.0886 > 0$). Model B selected ($\lambda_{\text{driven}} = 0.0426$).
6. Stage 3: Manifold QA	QA 50-step sweep along $D(\mathbf{W}_{in}) \in [0.01, 0.20]$	Verified cubic polynomial struct SmoothRidgeManifold	QA PASSED. $\lambda_{\text{driven}} = -0.0148 < 0$ everywhere. Mean $MC = 28.99$.
7. Stage 4: Continuous Bridge	Ingest 2AFC reversal task (16,029 human trials)	Formulated 5D Mossy Fiber vector $\mathbf{u}_t \in \mathbb{R}^5$ ($\Delta t = 10\text{ms}$)	Continuous temporal array padding for discrete events ($ttp \rightarrow ttr \rightarrow ttF$).
8. Stage 5: Cognitive Baselines	Fit discrete models (128 human participants)	Counterfactual EV-RW vs. Win-Stay / Lose-Shift (WSLS)	WSLS won baseline: Mean NLL = 53.25 (vs EV-RW 58.66), Total AIC = 14,144.9.
9. Stage 6: LOOCV & Halt	LOOCV out-of-sample choice predictive benchmark	Evaluated Unpatched Reservoir against WSLS baseline	Halt Triggered: Reservoir NLL 77.66 > 53.25 WSLS. Formulated 3 Patches.
10. Re-Evaluated Pure Choice	Re-run LOOCV on Patched Model (No DDM)	Applied Patches 1-3 (Flush, Pareto Tau, Asymmetric Plasticity)	Patched Choice NLL = 116.99, PR-AUC = 0.3856. Flush wipes magnitude state; Pareto causes autocorrelational lag.
11. Stage 7: Kinematics & DDM	Simultaneous Choice & Reaction Time prediction	Latency Readout RT_t + WSLS-DDM Drift Diffusion Bridge	WSLS-DDM Joint NLL = 20,702.2; Patched Reservoir Joint NLL = 33,056.8. Continuous latency tracking.

2 Detailed Analysis of Core Stage Interventions

2.1 Unbounded Logit Transformation & Boundary Elimination

The initial manifold optimization suffered from steep boundary clamping (BCR = 60%) near physical limits, causing optimizer instability and step discontinuities ($R^2_{\text{cubic}} = 0.7817$).

By transforming parameters into an unbounded latent space $\tilde{\Theta} \in \mathbb{R}^5$ using:

$$\tilde{\Theta}_k = \ln \left(\frac{\Theta_k - \Theta_{\min,k}}{\Theta_{\max,k} - \Theta_k} \right) \iff \Theta_k = \Theta_{\min,k} + \frac{\Theta_{\max,k} - \Theta_{\min,k}}{1 + \exp(-\tilde{\Theta}_k)} \quad (1)$$

we achieved ****analytical BCR = 0.0000 (0% boundary clamping)****, producing continuous polynomial trajectories across $D(\mathbf{W}_{in}) \in [0.01, 0.20]$ and guaranteeing spectral radius sub-criticality ($\lambda_{fb}(0.20) = 0.9783 \leq 0.99$).

2.2 C++ Biophysics & Tsodyks-Markram Short-Term Depression (D_t)

Un-normalized Mossy Fiber signals containing massive DC offsets (+10.0) drive standard reservoirs into severe tanh saturation (70.13% of neurons at $|x| > 0.90$), collapsing state variance to 0.7832.

Integrating the Tsodyks-Markram differential equation:

$$\frac{dD_t}{dt} = \frac{1 - D_t}{\tau_{\text{rec}}} - U_{\text{SE}} \cdot D_t \cdot u(t) \quad (2)$$

dynamically depresses effective input gain ($D_{\text{ss}} \approx 0.0062$), controlling pre-activation ($|h_{\text{pre}}| \leq 10.93$), eliminating overshoot saturation (**0.00% saturation**), and preserving rich linear dynamics ($\text{Var}[x_i] = 0.5636$).

2.3 Stage 6 & 7 LOOCV Choice-Latency Benchmarking

Evaluated across 128 human participants (16,029 trials):

- **Discrete Choice Prediction:** The discrete Win-Stay / Lose-Shift (WSLS) Markov baseline achieved the optimal out-of-sample fit (NLL = 53.25, PR-AUC = 0.6840). The unpatched continuous reservoir achieved NLL = 77.66, while the patched reservoir achieved NLL = 116.99.
- **Continuous Reaction Time Latency:** Bridging WSLS to a Drift Diffusion Model (WSLS-DDM) yielded a continuous Reaction Time RMSE of 0.5859 s and a total Joint Log-Likelihood of 20,702.2.

3 Computational Neuroscience Conclusions

1. ****Domain Specialization**:** Discrete Markov models (WSLS) are structurally optimal for discrete 2AFC choice switching tasks. 2. ****Reservoir Medium Strengths**:** Continuous-time reservoirs excel at continuous spatiotemporal motor control, dynamic memory integration, and multi-modal sensory mapping. 3. ****Biophysical Protection**:** Tsodyks-Markram Short-Term Depression (D_t) is mandatory for preventing non-autonomous saturation under un-normalized real-world driving.